Letian Yang

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EDUCATION

University of Southern California - Los Angeles, CA, U.S. Aug.2023-present
Ph.D. student, Department of Data Science and Operations (Statistics); Advisor: Prof. Dennis Shen

Southern University of Science and Technology (SUSTech) - Shenzhen, Guangdong, China Aug.2019-Jun.2023
B.S., Department of Statistics and Data Science, Major in Statistics; GPA: 3.78/4.00
Honors:
- National Scholarship (0.2%, Ministry of Education, China), 2022
- First Class of the Merit Student Scholarship for Exceptional Performance of SUSTech (5%), 2021&2020

University of California, Berkeley - Berkeley, CA, U.S. Jan.2022-Jul.2022
Exchange Student, Berkeley International Study Program; GPA: 4.00/4.00

RESEARCH INTEREST

High-dimensional statistics; causal inference; machine learning; deep learning.

ACADEMIC PEOJECT

Algebraic and Statistical Properties of the Partially Regularized OLS Interpolator
Advisor: Prof. Dennis Shen; manuscript submitted to AISTATS2025
Enhance the understanding of the overparameterized linear regression with the partially regularized OLS estimator and complements ongoing research on benign overfitting.
- Derived high-dimensional, partially regularized counterparts of Cochran's formula and the leave-one-out formula, both of which are agnostic to statistical modeling assumptions.
- Extended homoscedastic variance estimation under the Gauss-Markov model to the high-dimensional partial regression setting, providing a basis for conducting statistical inference, albeit with slight conservatism.

Refining Tissue-Specific Gene-Biological Process Relations with DNN, University of California Berkeley
Advisor: Prof. Haiyan Huang; ongoing
Develop a novel procedure to refine Gene Ontology (GO) annotations in a biological and tissue-specific context using transcriptomic data and existing GO annotations. The procedure incorporates deep neural networks (DNN), clustering, and stability testing to enhance gene function analysis across different tissues within specific biological processes.

PROFESSIONAL EXPERIENCE

Data Analysis Internship, Peking University Shenzhen Hospital Jun.2021-Aug.2021
- TCGA and GEO database analysis (Head and neck tumors)
  - Responsible for data mining and cleaning and implemented survival analysis, and prognostic factor analysis.
  - Conducted single gene expression analysis and found 60 genes for further study.
- Survey on oral health knowledge and behavior of dental patients
  - Responsible for questionnaire design, collection, and data processing.
  - Explored the differences in oral health and habits of people of different ages (especially children under 15), which helped doctors tailor the care plan.

ADDITIONAL INFORMATION

Programming Languages: R, Python, Java, MATLAB, LaTeX.
Systems and Tools: Linux, SSH, anaconda, AWS
Language: English (Fluent); Mandarin (Native Speaker).